

[Aashay Jadhav]

[CSE AIML]

Company: Nomura

Job Profile: CDO (Data Engineer/ Data Science Analyst)

Offer Type: [Internship Only / Placement Only / Internship + Placement]

Internship Stipend: 75000

CTC:13.66 (2L signing Bonus + 11.66 Base)

Location: Powai - Mumbai

Process Date: 28 July PPT - 31 July OA - 1 August Interview

Placement journey: The placement journey began in the last week of June when JPMC first visited us. My first Target company was MSCI, I had prepared towards it for almost 2 months, Got shortlisted in the OA, then cleared the first tech interview, then the second, I could see many good candidates getting rejected while I got till the hr round, felt today is my day, but it wasn't. Got rejected after the hr round. Took me almost 2-3 days to get back to studying as usual. Now I did have 3 preferences for a company 1: Should have a Data and AI related role, 2: Should be in Mumbai, 3: CTC should ideally be somewhere around 12. Which is why in the first 10 companies, I was only interested in MSCI. For other companies I did give the OA to make sure I get some practice of time bounded problem solving, but solved only enough to not get shortlisted. Then IDFC came, again cleared the OA, the first round did not go well, still got shortlisted for second round and then that 2nd round was my worst ever. Interviewer was asking me questions I had no idea about, (software testing, software engineering, deployment) I had no option but to keep on saying that sir I don't know. So I got rejected. Then came Nomura...

Placement Process:

Nomura visited for 3 roles:

ITD

CDO:

https://docs.google.com/document/d/1ntYAHYcKOkuCpI2mwIhOWmvhz4NgVfL2YF2yuLjhwWU/edit?usp=drive_link

GM

Criteria for ITD was OA Aptitude while for others it was just resume shortlisting. We were asked to select 2 role preferences. But I think that OA score was also considered for other 2 as I got shortlisted for all 3 roles.

1. **[OA]** [200]

This was a complete Aptitude test, consisting of 3 sections: Verbal, Logical Reasoning, Quants Aptitude. For all three I had practiced from IndiaBix. Since I had given almost 10 OA's by now, I was confident (this really matters). The day before I gave practice tests for each of these sections on IndiaBix and realised that I was scoring maximum in Quants apti, followed by logical and then verbal. So in OA I began with Quants then logical then verbal, although the sections were in opposite ordering. Many students followed the given

(verbal-logical-quant) order and thus had almost no time for many quant questions. So identify your strong area and begin with that.

I practiced only from IndiaBix for this, I began practicing for Aptitude probably in July itself, focused more on Aptitude after MSCI.

Interview Day:

I was in all the 3 shortlists, while this gave me more chances, there was a possibility of me losing focus, so ensure that you cut out all the noise and stay focused on the day of the interview. Now first the GM GD happened, topic- Is AI making us smarter or dependent, this did not go well for me and I did not go further. Then for ITD something unexpected happened, Due to suspicion of copying, all 28 candidates were asked to give another pen-paper OA, where they just shortlisted 7 for further rounds(no one knows what criteria was followed as most of us had solved all the questions correctly). So I did not get through in ITD as well. By the time my CDO Interview began I was already out of the other two.

2. **[Tech Interview]** [8-CDO, 28-ITD, 19-GM]

So the interview began with them asking me to skip introduction and just give one line description of my projects and my role in them. I had 4 projects mentioned, all related to Data Analysis, ML and LLM, GEN AI. So my resume was highly aligned with the role. I had done an internship, so I also explained about my role in that and spoke about a few of my skills. There were 2 interviewers. One of them started questioning about one of my (Gen AI powered tool, that converts wikipedia topics into easy to understand comic strips) project. She first asked who is your target audience for this project. I answered that the younger students in age 12-20 who are inquisitive to learn about new things. Then she asked how will you control younger students from generating images of vulgar content. I tried explaining that firstly, wikipedia won't have much vulgar content, and secondly we were taking user's age as one of the user preferences and were passing it through prompts for our model. So we can ensure via structured prompting that the user does not get any vulgar images. She honestly was not convinced. The other interviewer then began by giving a problem. I have 1000 resumes, and want to perform searching in this, how can I do it(similar question was asked in IDFC as well), I explained the entire RAG framework and how it would make the searching very easy. Then he gave me another problem that, every morning he has to get up at 4 and mark a set of tasks as 0 or 1 based on the result, how would I solve it. Now just when he began writing 0 and 1 Logistic regression clicked in my mind. I explained to him the entire flow and working of logistic regression. He was satisfied with both my answers. Then he asked more about what I had done in internship. I spoke about flight price prediction (regression) model. I had mentioned R2 score as an accuracy metric in the resume. He questioned that we don't use R2 score to measure accuracy, I told him that since it was a regression model we can't use accuracy/precision, etc and rather we use mse/rmse/r2 score. He then asked me to explain what R2 score is. I knew about it and explained him with a short example. He then asked me to explain precision and recall. I answered that correctly as well. Then he asked me to give one line explanation about Encoder and Decoder and their examples. I gave a satisfactory answer to that as well. Now in the meantime, the other interviewer had left the room to take a call, this can be a bit distracting, but just try to focus on your answers. Also whenever interviewer asks u about a question, try to question him back, to get complete understanding of his question if you did not understand something. He then asked me to mention all ML algos I know about. I mentioned linear reg, logistic reg, random forest, decision trees, etc. He asked whether I knew about XGBoost. I told him that I had idea about its working but did not have experience working with it. He gave a remark that, did you just move to learning LLMS without completing all ML algos? I mentioned that I knew about them but had not worked with all the ML algos. He then asked me, how good am I in

python, I said 8 on 10. He then asked a DSA question, given an array of numbers find the minimum elements you need to choose to reach the target value. I thought for about 2-3 minutes,(greedy, dp,etc) but could not get an ans, then i just thought of using the brute force and told him for each element we will choose it and then check how many more we need to reach target and maintain a minimum count. So accepted the brute force (since this role did not require much dsa knowledge). Then they asked 2 sql questions 1 was a group by and count query (given document id and curr_status, count the no. of days that document is in the curr_status, each row denotes a day). Then next she gave 2 columns and asked me to explain all types of joins. They then asked if I had any questions for them. I did ask one about gen ai applications in nomura. And then about the diff in culture that nomura has as compared to other companies. They answered these for almost 5 mins.

Overall, the interview went well, I was expecting to be called for next round.

3. **[HR]** [2- CDO]

Only 2 candidates gave this round for CDO. It was a casual round. Interviewer asked me to give an intro, then asked me about the field I want to work in. Then she said that since I was shortlisted for all 3 roles, give me an explanation about all 3 roles. Then family background and higher studies preference(definitely no). And then I asked her 3 questions about what do u look for in a person not tech skills but general skills, one good thing about nomura and one not so good thing. She answered well for 5-10 mins.

But then she left the room with me and when we reached the tpc standing outside tpo, she just said he is done for the day, he can go home. I was like what?? Nothing went alarmingly wrong in the interview, why am I being asked to go home, the tpc also said that you are not selected so you can go home now. I felt, I will have to start preparing for companies again from tomorrow. So both the hr round candidates were asked to go home. Then message came that every cdo candidate can go home and results will come later. So then there was some hope, given that only 2 gave the hr round. And finally the result came.

2 were shortlisted for CDO, 1 FOR GM, 3 FOR ITD.

Useful Tips:

Resume:  **Aashay_Resume**

Leetcode: https://leetcode.com/u/Aashay_27/

Resources:

Java for coding: <https://www.youtube.com/playlist?list=PLfqMhTWNBT3LtFWcwpqTkUSIB32kJop>

Java for interview: <https://www.geeksforgeeks.org/java/java-interview-questions/>

Collections framework: <https://www.youtube.com/watch?v=rzA7UJ-hQn4>

OS for interview: Gatesmashers and

<https://www.geeksforgeeks.org/operating-systems/operating-systems-interview-questions/>

Dbms for int: Gatesmashers and

<https://www.geeksforgeeks.org/dbms/commonly-asked-dbms-interview-questions/>

Ccn: i practiced from gpt, for(os,ccn,dbms i followed college curriculum topics, not all but only imp ones).

General Imp Question:

If you have AI in your resume learn about RAG, as interviewers generally know that in AI (asked in all 3 companies for me).

HashMap, HashSet, ArrayList internal implementation in java, JRE/JDK/JVM/JIT in java.

All types of questions on Palindrome(easy to hard), and sorting algos.

Indexing in dbms, have strong hold on sql mostly joins and subqueries are asked.

My Journey:

I completed an internship and a Data Analysis course after second year. Won a few DA competitions in November, but still in December I was really not confident of having a chance at getting a good placement, I did not have great dsa knowledge either. I effectively began my journey in first week of January. Earlier I used to practice very few dsa problems and that too in python. But my seniors told me about how important java is for companies so I switched to java. (I did it because I was not too advanced in dsa then and was anyways going to start from scratch.) If you are solving in python/c++ pls continue, just do oops and java workings well theoretically.

January: began with java, close to 80 dsa problems, 1 hack participation

February: Gen AI course,close to 50-60 dsa problems

March: Gen AI course, 100 dsa problems, began trying lc contests

April: 150 dsa problems, completed the course, finalist in hackathon(first time)

May : 50-60 dsa, won here technologies hack(first time)

June: Did not focus on solving new problems, rather kept revising topics and solved problems, began with theory topics - java,oops,dbms,os,ccn

Resume: Do not lie about anything, first interview rounds are completely based on your resume. If you dont know about anything that is written it gives a bad impression. Know about every project in detail. Its ok to not know a lot of things, just know whatever you have written. My resume was completely AI/ML/DA/LLM/GENAI based, absolutely nothing about development as I did not know much about it.Also mention those things at the top which you want the interviewer to see, there is no specific order. 3-4 projects are enough, some skills. Pls do mention about any sports you would have played in your life. My MSCI and IDFC first round interviewers were very interested in my cricket skills and that really helped me. Just add one line but do add.

Projects: You don't need to have the best projects, you can have okish projects but need to know everything you did. You can take them from github, but again, know everything. If you can deploy them somewhere, that can be good as you can show it live to interviewer. But that is absolutely not necessary.

Interview: If you are asked something you don't know about, there are 2 ways, either you have absolutely no idea about it- just say sir i am not aware about it. Or you can relate it to some other topic. In that case mention that i don't know exactly, but i feel this is similar to the other thing i have worked on. First round is just resume based, no need to worry much. Also just to impress the interviewer don't speak fancy words which you are not aware of or have good knowledge about. It happened with me at MSCI, where I mentioned an advanced topic too early, so then he went a bit deeper into it and I was not able to answer all of it. The interviewers are mostly chill so they will calm you down.

DSA: DSA is necessary, but 500-700 problems are not, you can crack the best of companies even with less problems solved. But the more you solve, the more confidence you will have of solving unseen problems in oa or interview. Confidence is really important in dsa as in the pressure if you can't think of the right approach, it can get difficult. Codeforces, CP is absolutely **not needed** for any company other than PHONE PE. I still haven't solved any problem on codeforces. If you like solving that's great but please ensure you are working on other aspects also. Also do ensure that you can take input in your language, as leetcode does not prepare us for that. Leetcode contests solving can be helpful, to get practice but don't panic if you can solve only 1 or two problems or even if you can't solve any problem in a contest. You can directly skip HARD problems on LC. Follow strivers A2Z Sheet. MSCI, Barclays and UBS had medium problems but idfc had easy problems. In interview mostly easy -medium are asked or they may not even ask. Even if you have not solved many problems, in June try to read through all easy-medium problems in strivers sheet. Having an idea can help you at least come up with an approach in interviews.

OS/CCN/JAVA OOPS/DBMS/Aptitude/puzzles: I began with all this in June, You can start in May ideally. Java oops are extremely important, dbms comes next, along with sql, os is slightly important, while ccn mostly won't be asked, although it's good to know basics.

Hackathon: Please participate in hacks. If you don't have the confidence, or no one is taking you in their team, find 3 others like you and participate. Because A- it gives you some impactful projects and B- if you sincerely give your best for 24 hours, for 4-5 hacks you will eventually win one for sure. It is helpful in interviews as it validates your skills for the interviewer. Also winning hacks don't require extraordinary skills these days. Just you need to be able to use open source tools well and present well. Please participate in company hackathons, if you win or even reach finale, they will select you for placement. Although if you are not able to win, that is not the end of the road, it's not at all necessary.

Role: If you have certain preferences it's ok to let go of some roles. But please be realistic, if you have confidence in your preparation, only then leave companies. Because it may happen that good companies might go away. Also please don't be too dependent on one role/company. MANY PEOPLE WERE WAITING FOR WORKINDIA DA, BUT IT DID NOT COME. At the same time it is important to have a few target companies and give it your all for it.

I won't say that placement is very much dependent on LUCK, It plays some part, but if you have not prepared well you have very little chance that luck would get you placed. But if you are well prepared luck can take you that extra mile. Overall if you have prepared well, You will do well eventually.

If you have given your best effort right from the time you decided to be serious (6/5/4/3/2 months) something good will happen, just keep going, Rejections can be tough, but eventually it will help you get something better. In my case I did not get MSCI, but rather got my preferred role, location and pay.

ALL THE BEST!!

Dont hesitate to reach out, 9372138292